

CS 736: Medical Image Computing

(Instructor : Suyash P. Awate)

Mid-Semester Examination

Date: 25 Feb 2021

Duration: 2 hours

Roll Number: _____

Name: _____

When in doubt, make suitable assumptions, state them clearly, and proceed to solve the problem.
All the best!

Formulae for Reference

Poisson probability mass function: $P(x; \lambda) := \lambda^x \exp(-\lambda)/x!$ with mean = λ , variance = λ , and mode = $\text{floor}(\lambda)$.

Rician probability density function: $P(x; \nu, \sigma) := \frac{x}{\sigma^2} \exp(-\frac{x^2 + \nu^2}{2\sigma^2}) I_0(\frac{x\nu}{\sigma^2}), \forall x \geq 0$

Gamma probability mass function: $P(x; \alpha, \beta) := \eta(\alpha, \beta) x^{\alpha-1} \exp(-\beta x)$, where $\eta(\alpha, \beta)$ is a normalization constant that is a function of parameters α and β . In this case, mean = α/β , variance = α/β^2 , and mode = $(\alpha - 1)/\beta$.

Rayleigh probability density function: $P(x; \sigma) := \frac{x}{\sigma^2} \exp(-\frac{x^2}{2\sigma^2}), \forall x \geq 0$

1. [20 points]

- (5 points: 2 + 3) Formulate the k-means clustering algorithm as an optimization problem. Specify an implementable algorithm for solving the optimization problem.
- (5 points) Specify an implementable algorithm relying on principles in simulated annealing to solve the optimization problem $\min_{x,y} f(x,y)$, where the function $f(\cdot)$ is real valued, variable x is a multivariate continuous, and variable y is multivariate discrete.
- (10 points) Specify an implementable algorithm to extend the k-means clustering algorithm by relying on principles in simulated annealing.

2. [10 points]

Consider an extension of the k-means clustering algorithm where the “means” are constrained to be among one of the data points. Formulate an optimization problem and specify an implementable algorithm for this extension of k-means.

3. [30 points]

Consider an acquired real-valued noisy image Y . Consider the noise model to be independent and identically distributed (i.i.d.) zero-mean additive Gaussian.

Consider that, in the original real-valued noiseless image X , every pixel had one of only two possible intensities, say, $\alpha_1 \in \mathbb{R}$ that models the intensity of objects and $\alpha_2 \in \mathbb{R}$ that models the intensity of the background. We know that the intensities in the noiseless image X are spatially piecewise constant. At pixel i , let the noiseless intensity X_i equal α_1 with probability U_i ; and let X_i equal α_2 with probability $(1 - U_i)$.

You want to formulate an optimization problem to solve for the unknown variables α_1, α_2 and the probabilities $\{U_i\}_{i=1}^N$ at the N pixels in the image, taking into account ALL the aforementioned information.

- (5 points) Design a Markov-random-field based prior model on the image U of probabilities $\{U_i\}_{i=1}^N$. Specify a neighborhood system, the cliques, and the potential functions.
 - (10 points) Design a data-fidelity based energy/loss function on the image U and intensities α_1 and α_2 .
 - (5 points) Formulate the optimization problem.
 - (10 points) Specify an algorithm for solving the optimization problem formulated above, including update formulae for α_1, α_2 , and $\{U_i\}_{i=1}^N$.
4. [15 points] All the following questions here pertain to magnetic resonance imaging. Consider a homogeneous tissue that isn't magnetic when it exists without the influence of any external magnetic field.
- For each of the following questions, state what happens to the aggregate tissue magnetization vector $\langle m_x, m_y, m_z \rangle$ when the tissue is placed in the static external magnetic field $\langle 0, 0, B_0 \rangle$, i.e., specify the following:
- (i) the steady state, i.e., $\langle m_x(t = \infty), m_y(t = \infty), m_z(t = \infty) \rangle$;
 - (ii) the magnitude of the XY-plane component at some finite time $0 < t < \infty$ (mathematical equations aren't needed; plotting a graph is sufficient);
 - (iii) the magnitude of the Z-axis component of the magnetization at some finite time $0 < t < \infty$ (mathematical equations aren't needed; plotting a graph is sufficient);
 - (iv) precession frequency; and
 - (v) if precession will be observed in the specific scenario mentioned.

- (5 points) Consider that, at time zero, the aggregate magnetization is: $m_x(t = 0) = m_y(t = 0) = m_z(t = 0) = 0$.
- (5 points) Consider that, at time zero, the aggregate magnetization is: $m_x(t = 0) = m_y(t = 0) = \alpha \gg 0$ and $m_z(t = 0) = 0$.
- (5 points) In the previous question, for two different homogeneous tissues, state the differences.

5. [20 points]

Consider a dictionary learning problem where, given a dataset $\{x_i \in \mathbb{R}^2\}_{i=1}^N$ with $N = 1000$, the goal is to learn the dictionary elements/atoms $\{d_j \in \mathbb{R}^2\}_{j=1}^M$, where $M = 3$, by solving via the L_0 pseudo-norm regularization enforcing sparsity on the dictionary coefficients.

For each of the following questions, assume that the objective-function value at the global minimum turns out to be zero, i.e. the dictionary-based model fits each datum perfectly under the given constraints. Thus, in that case, state what is the implied data distribution by drawing a clear picture and explaining briefly in words.

- (5 points) Consider the sparsity constraint that allows only a single atom to be used in the sparse representation of each data point x_i .
- (5 points) Consider the sparsity constraint that allows only a single atom to be used in the sparse representation of each data point x_i . At the same time, consider that coefficients are constrained (hard) to be non-negative.
- (5 points) Consider the sparsity constraint that allows at most two atoms to be used in the sparse representation of each data point x_i . At the same time, consider that coefficients are constrained (hard) to be non-negative.
- (5 points) Consider the sparsity constraint that allows at most two atoms to be used in the sparse representation of each data point x_i . At the same time, consider that coefficients are constrained (hard) to be non-negative and constrained to sum to one.